

Saturated Superfluid Vacuum Goldstone

The Photon as a Goldstone Mode and the Chiral-Shear Near Field

Stig Norland
Independent Researcher
Bergen, Norway

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Abstract

This paper isolates the carrier problem in the electromagnetic sector of the Saturated Superfluid Vacuum (SSV) series. Static Coulomb and magnetic near-field structure arises from chiral-shear flow gradients around toroidal charged vortices, with coupling strength set by the dimensionless parameter α . Identifying the propagating photon with a chiral-shear mode does not work: linearising the full LogSE + chiral-shear + CP¹ functional around the uniform vacuum shows the chiral-shear term is *silent* in the linear spectrum — the linear-order current is a pure gradient and the $|\nabla \times \mathbf{j}|^2$ energy is quartic — so no propagating chiral-shear mode exists at any speed, and the internal-direction channel disperses as $\omega = \hbar k^2/2m_0$: gapless, quadratic, charge-decoupled at linear order.

Eliminating that candidate does not establish the alternative. The saturated order parameter has a longitudinal Goldstone/Bogoliubov sound mode at c , but a scalar phase mode alone is not a photon: it supplies neither two transverse polarizations nor Maxwell gauge structure, and the CP¹ direction is charge-decoupled at linear order, so its transport cannot simply be asserted to provide them. The chiral-shear sector remains physically real and is assigned to static and near-field structure — Coulomb pressure, magnetic circulation, Aharonov–Bohm phase, vortex-ring self-energy.

The task is therefore a two-channel derivation: obtain the radiative Maxwell sector from the Goldstone phase channel at speed c while preserving the static Coulomb and magnetic results as the near-field limit of chiral-shear topology. That derivation is **not** claimed here. This paper states the consistency conditions, identifies the matching problem between the channels, and fixes the success criterion: a transverse, gauge-like two-polarization sector with displacement-current coefficient $1/c^2$.

Contents

1	Introduction	3
2	The Problem in Paper II	3
3	Goldstone Channel of the Saturated Medium	4
4	Two-Channel Electromagnetism	4
5	What Must Survive	5
6	The Matching Calculation	6
7	Avoiding New Circularity	6
8	Consequences	7
9	Discussion	7
10	Conclusion	7

1 Introduction

The electromagnetic sector of SSV has split into two logically distinct pieces:

- (i) a *static near-field sector*, where charge, Coulomb force, magnetic circulation, and AB phase arise from the topology and chiral-shear flow of toroidal vortices;
- (ii) a *radiative sector*, where the observed photon must propagate at the measured vacuum speed c .

Paper II developed the first piece but used the wrong carrier for the second. It treated the chiral-shear mode as the photon and assigned it the speed $c_\perp$. This cannot be the final physical interpretation because the observed photon propagates at c .

Analogue-gravity and superfluid-vacuum literature motivates emergent collective fields [1], while Zloshchastiev [2] supplies the logarithmic wave equation. Neither cited source establishes the SSV-specific identification of the physical photon with the scalar phase mode of Ψ . That phase mode propagates at the medium sound speed,

$$c_s(\rho_0) = c, \quad (1)$$

and is therefore compatible with observed light propagation and with gravitational-wave constraints such as GW170817.

The question is not simply whether one can rename the photon. The hard question is whether the successful static electromagnetic mechanisms of Paper II survive the replacement:

$$\text{chiral-shear photon at } c_\perp \longrightarrow \text{Goldstone photon at } c.$$

The present paper formulates that problem.

Candidate programme

The Goldstone/Bogoliubov mode supplies a candidate carrier speed c , while the chiral-shear field supplies a candidate static near-field sector. A photon identification requires a derivation of two transverse polarizations, Maxwell dynamics and source matching; none follows from the scalar sound mode alone.

2 The Problem in Paper II

Paper II recovered a set of electromagnetic structures from chiral vortex topology:

$$F_C(r) = \frac{\alpha \hbar c}{r^2}, \quad (2)$$

$$\mathbf{B} \sim \nabla \times \mathbf{v}_\perp, \quad (3)$$

$$\gamma_{AB} = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\mathbf{l}. \quad (4)$$

Here $\mathbf{v}_\perp$ is the chiral-shear flow associated with the toroidal vortex, and α is the empirical dimensionless strength of that sector.

The same section then tentatively identified the propagating photon with a transverse chiral-shear phonon satisfying

$$\partial_t^2 \delta\theta = c_\perp^2 \nabla^2 \delta\theta, \quad c_\perp \neq c. \quad (5)$$

This produces the speed problem. If the physical photon propagated at $c_\perp$, vacuum light speed would be $c_\perp$, not c . The earlier patch that $\alpha \rightarrow 1$ in free vacuum is not available inside SSV because vacuum *is* the saturated medium.

The diagnosis is therefore:

$$\text{static chiral-shear mechanism: keep,} \quad \text{radiative photon identification: replace.} \quad (6)$$

The SSV issue 138 computation subsequently showed that equation (5) was not merely the wrong photon — it does not arise from the functional at all: around the uniform vacuum the chiral-shear term contributes nothing at linear order, and the internal-direction channel it was thought to describe is first-order in time with $\omega = \hbar k^2/2m_0$. The “slow second light” has no linear mode behind it; the diagnosis above is thereby upgraded from an audit conclusion to a computed result.

3 Goldstone Channel of the Saturated Medium

Write the order parameter as

$$\Psi(\mathbf{x}, t) = \left(\sqrt{\rho_0} + \frac{\delta\rho}{2\sqrt{\rho_0}} \right) e^{i\theta_0 + i\delta\theta}. \quad (7)$$

Linearisation gives coupled density and phase perturbations:

- $\delta\rho$ is the density component;
- $\delta\theta$ is the phase component.

Together they form the longitudinal Bogoliubov sound branch. At long wavelengths its phase component obeys

$$\partial_t^2 \delta\theta = c^2 \nabla^2 \delta\theta, \quad (8)$$

with dispersion

$$\omega = ck. \quad (9)$$

This branch has the required propagation speed, but it is a scalar longitudinal sound mode. Speed agreement is necessary and not sufficient for identifying it with physical electromagnetic radiation.

Candidate channel assignment

The Goldstone/Bogoliubov branch is the only current candidate carrier at c . The chiral-shear channel is a candidate near-field structure of charged defects. Their electromagnetic interpretation remains unproved.

4 Two-Channel Electromagnetism

The corrected EM sector should be written as a two-channel theory:

$$\text{near field:} \quad \mathbf{v}_\perp, \Phi_\perp, \mathbf{A}_\perp, \quad \text{chiral-shear/topological sector,} \quad (10)$$

$$\text{radiation:} \quad \delta\theta_G, \quad \text{Goldstone phase channel.} \quad (11)$$

The near field knows about charge chirality and the dimensionless coupling α . The radiative field knows about propagation at c .

This is not unusual from the point of view of ordinary electrodynamics. In QED, the Coulomb field is not a freely propagating transverse photon; it is the constrained or longitudinal part of

the gauge field. Radiation is carried by transverse on-shell photon modes. SSV should reproduce this division mechanically:

$$\text{Coulomb interaction} \leftrightarrow \text{static chiral-shear pressure constraint}, \quad (12)$$

$$\text{on-shell light} \leftrightarrow \text{Goldstone phase wave at } c. \quad (13)$$

The task is to derive the effective Maxwell system in which the source sector is chiral-shear/topological, while the homogeneous radiative sector propagates at c :

$$\nabla \cdot \mathbf{E} = \rho_{\text{ch}}/\epsilon_0, \quad (14)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (15)$$

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad (16)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2} \partial_t \mathbf{E}. \quad (17)$$

Relative to the provisional Paper II derivation, the displacement-current coefficient must be $1/c^2$ rather than $1/c_\perp^2$.

5 What Must Survive

The Goldstone correction is acceptable only if it preserves the successful static and topological content of Paper II. The following structures must survive.

Coulomb strength. The static force must remain

$$F_C(r) = \frac{\alpha \hbar c}{r^2}. \quad (18)$$

In the corrected reading, α is not a photon-speed ratio. It is the strength of the chiral-shear near-field coupling.

Magnetic topology. The magnetic field must remain the curl of a vector flow,

$$\mathbf{B} \propto \nabla \times \mathbf{v}_\perp, \quad (19)$$

so that $\nabla \cdot \mathbf{B} = 0$ remains a vector identity and magnetic monopoles remain topologically forbidden.

Aharonov–Bohm phase. The phase shift must remain a Berry phase of the charged ring in a background circulation:

$$\gamma_{\text{AB}} = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\mathbf{l}. \quad (20)$$

This is naturally near-field/topological rather than radiative.

Anomalous magnetic moment. The one-loop self-energy picture must be reinterpreted carefully. The virtual exchange responsible for $g - 2$ may continue to involve the chiral-shear near field, but the on-shell photon propagator must belong to the Goldstone channel. The calculation therefore has to distinguish virtual constrained exchange from physical radiative quanta.

Open matching problem

The central technical challenge is to show that the chiral-shear near field and the Goldstone radiation field are not two unrelated electromagnetic fields, but two limits of one effective EM response of the saturated medium.

6 The Matching Calculation

A complete derivation should start from the full SSV functional,

$$S[\Psi] = \int dt d^3x [\mathcal{L}_{\text{LogSE}}(\Psi) + \mathcal{L}_{\perp}(\Psi, \nabla\Psi) + \mathcal{L}_{\text{defect}}], \quad (21)$$

expand around a charged toroidal background Ψ_e , and separate fluctuations into constrained near-field and propagating Goldstone parts:

$$\Psi = \Psi_e + \delta\Psi_{\text{near}} + \delta\Psi_{\text{rad}}. \quad (22)$$

The desired effective action has the schematic form

$$S_{\text{eff}} = \int dt d^3x \left[\frac{\epsilon_0}{2} E^2 - \frac{1}{2\mu_0} B^2 + J^\mu A_\mu \right], \quad (23)$$

with two requirements:

$$\begin{aligned} \text{static limit: } \nabla^2\Phi &= -\rho_{\text{ch}}/\epsilon_0, & F_C &= \alpha\hbar c/r^2, \\ \text{radiative limit: } \partial_t^2 A_i - c^2 \nabla^2 A_i &= 0, & \omega &= ck. \end{aligned} \quad (24)$$

The calculation is therefore a matched asymptotic problem. It must test whether chiral-shear topology can supply the source and static response and whether a far-field disturbance can project onto a transverse sector transported at the Goldstone-branch speed.

7 Avoiding New Circularity

The Goldstone photon correction must not simply insert Maxwell theory by hand. The following shortcuts would not count as derivations:

- (i) postulating a separate electromagnetic gauge field in addition to Ψ ;
- (ii) replacing $c_{\perp}$ by c in Paper II's equations without deriving the channel projection;
- (iii) keeping Coulomb from chiral shear and radiation from Goldstone without deriving their matching at sources;
- (iv) using observed light speed to tune a coefficient in the chiral-shear sector.

A valid derivation should show that the same saturated order parameter supports both the constrained static response and the propagating radiation response, with no additional gauge field inserted by hand.

Success criterion

The Goldstone photon programme succeeds if the full SSV linearisation around charged toroidal defects yields Maxwell electrodynamics with static Coulomb strength $\alpha\hbar c$, magnetic topology from chiral-shear circulation, and on-shell radiation satisfying $\omega = ck$.

8 Consequences

If the matching calculation succeeds, several tensions in Paper II are resolved at once.

Light and gravity share the correct speed. Both electromagnetic radiation and gravitational disturbances can be read as Goldstone/Bogoliubov-channel excitations at c , consistent with GW170817.

α is not a speed ratio. The fine-structure constant becomes a chiral-shear/topological coupling and defect-geometry parameter, not a ratio involving the observed photon speed. This matches the separate alpha-paper proposal $\alpha = \xi/R^*$.

The old EM derivation becomes a near-field derivation. The Paper II chiral-shear equations are not discarded; they are reclassified. They describe constrained near-field structure and virtual exchange rather than the on-shell photon.

Magnetic monopoles remain forbidden. Because the magnetic field still descends from a curl of the chiral-shear flow, $\nabla \cdot \mathbf{B} = 0$ remains topological rather than postulated.

9 Discussion

The Goldstone photon problem is a cleanup calculation, but it is not a minor one. It decides whether the SSV electromagnetic sector can be made internally coherent after the propagation-speed audit. The desired result is conceptually simple: sources live in chiral topology; radiation lives in the Goldstone phase channel. The hard part is proving that these are two faces of one effective electromagnetic field.

This also clarifies the status of Paper II. Its static Coulomb and magnetic results remain valuable because they identify how charge and magnetic topology arise mechanically. Its original radiative photon identification should be treated as provisional scaffolding. The present paper states the corrected target.

10 Conclusion

The current SSV functional rules out the former slow chiral-shear photon. Its remaining Goldstone/Bogoliubov sound branch propagates at c , but the paper has not derived a photon from it. The chiral-shear near-field interpretation is likewise conditional on the matching calculation.

The open derivation is the matching calculation:

$$\boxed{\text{chiral-shear near field} \longleftrightarrow \text{Goldstone radiation at } c}$$

If this can be derived from the full SSV functional without adding a separate gauge field, the electromagnetic sector becomes substantially more coherent. If it cannot, the SSV force-sector programme must be revised at one of its central joints.

References

- [1] G. E. Volovik, *The Universe in a Helium Droplet* (Oxford University Press, Oxford, 2003).
- [2] K. G. Zloshchastiev, “Superfluid vacuum theory and deformed dispersion relations,” *Int. J. Mod. Phys. A* **35**, 2040032 (2020); arXiv:2011.11897.